

Emily Simon

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University of Chicago

Education

University of Chicago <i>Ph.D. in Astrophysics</i>	<i>Sept 2021 – Present</i>
◦ Thesis: Multi-messenger signals of astrophysical shocks ◦ Advised by Damiano Caprioli	
Columbia University <i>B.A. in Astrophysics</i>	<i>Sept 2018 – May 2021</i>
◦ Advised by Lorenzo Sironi & Luca Comisso	
Borough of Manhattan Community College (CUNY) <i>56 credits toward A.S. in Physics</i>	<i>Sept 2016 – May 2018</i>

Awards and Fellowships

Eugene N. Parker Fellowship	<i>2026</i>
Illinois Space Grant Consortium Graduate Fellowship	<i>2025 - 2026</i>
NSF GRFP Fellowship	<i>2021 - 2026</i>
Elaine K. Bernstein Women in Science Award	<i>2021</i>
De Lara Scholarship, Columbia	<i>2020</i>
AstroFest 1st Prize Poster Award, Columbia	<i>2019</i>
Phi Theta Kappa Scholarship, Columbia	<i>2019</i>
BMCC Achiever Award for 4.0 GPA, CUNY	<i>2018</i>

Previous Research Positions

Undergraduate Research Assistant (high energy theory), Columbia	<i>2018-2021</i>
Summer Undergraduate Research Assistant, CERN, U. of Michigan, Columbia	<i>2020</i>
CUNY AstroCom Fellow, American Museum of Natural History & CUNY	<i>2017-2018</i>

Other Education: Workshops, Conferences, & Summer Schools

NSF Plasma Physics (GPAP) Summer School, Swarthmore	<i>2023</i>
Foundations of Cosmic Ray Astrophysics Summer School, International School of Physics “Enrico Fermi”, Varenna, Italy	<i>2022</i>
CERN REU Particle Physics Course, CERN & U Michigan	<i>2020</i>
CUWiP Princeton Plasma Physics Workshop, PPPL	<i>2019</i>

Publications

- E. Simon**, R. Diesing, D. Caprioli, S. Sclafani, *Detecting Neutrino Emission from Supernova Remnants: A Theoretically Motivated Target Catalog*, 2026b, [ApJ](#), **999**, 86 [🔗](#).
- E. Simon**, D. Caprioli, C. Haggerty, B. Reville, *Novel Setup for Hybrid Simulations of Astrophysical Shocks*, 2026a, [ApJ](#), **997**, 144 [🔗](#).
- K. Berghaus, D. Hooper, **E. Simon**, *Searching For Superheavy Decaying Particles With Ultra-High-Energy Neutrino Observatories*, 2025, [JCAP](#), **07**, 067 [🔗](#).
- E. Simon**, D. Caprioli, C. Haggerty, B. Reville, *Maximum Energy Achievable in Supernova Remnants: Self-Consistent Simulations*, ICRC Proceedings of Science, 2024, [arXiv:2402.01048](#) [🔗](#).
- A. Dekker, I. Holst, D. Hooper, G. Leone, **E. Simon**, H. Xiao, *Diffuse Ultra-High-Energy Gamma-Ray Emission From TeV Halos*, 2024, [PRD](#), **109**, 083026 [🔗](#).
- Q. Minor, M. Kaplinghat, T. H. Chan, **E. Simon**, *Inferring the concentration of dark matter subhaloes perturbing strongly lensed images*, 2021, [MNRAS](#), **507**, 1 [🔗](#)
- Almazan et al. (incl. **E. Simon**), *Search for non-pointing and delayed photons with ATLAS in pp collisions at $s = 13$ TeV*, ATLAS Internal Note, 2020
- Polzin et al. (incl. **E. Simon**), *Astronomy as a Field: A Guide for Aspiring Astrophysicists*, 2023, [arXiv:2312.04041](#) [🔗](#).

Contributed Talks & Posters

- Symposium on Lepton Photon Interactions at High Energies**, Madison, Wisconsin, USA (August 2025), *Selecting Hadronic Supernova Remnants for Targeted Galactic Neutrino Searches*
- International Conference for Cosmic Rays (ICRC)**, Geneva, Switzerland (July 2025), *Selecting Hadronic Supernova Remnants for Targeted Galactic Neutrino Searches*
- Supernova Remnants III**, Crete, Greece (June 2024), *Shock Obliquity in Hybrid Simulations of Supernova Remnants*
- International Conference for Cosmic Rays (ICRC)**, Nagoya, Japan (August 2023), *The Maximum Energy Achievable in Supernova Remnants: Hybrid Simulations*
- ATLAS Collaboration Summer REU Presentations**, Remote due to Covid-19 (August 2020), *Search for SUSY with delayed and non-pointing photons*
- AstroFest**, Columbia University (September 2019), Poster: *Magnetic Reconnection as a Potential Driver of Non-Thermal Particle Emission in Astrophysical Plasmas*
- American Astronomical Society 233rd Meeting**, Seattle, WA (January 2019), Poster: *Perturbations of Gravitational Lenses Due to Low-Mass Dark Matter Halos*

Teaching

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| Teaching Assistant , UChicago, Black Holes | 2026 |
| Teaching Assistant , UChicago, Stars | 2025 |
| Main Instructor , UChicago, Physics of Stars | Summer 2022-24 |
| ◦ Designed and taught an immersive three-week introductory pre-college astronomy class. | |
| Lead Tutor, Geometry & Calculus | 2019 - 2021 |
| ◦ Columbia University, Tutoring & Learning Center (TLC) | |
| Teaching Assistant , Columbia, Stars, Galaxies, & Cosmology | 2020 |

Selected Outreach & Service

Invited talk, NASA Community College Network (NCCN), 2026

- *What are gamma rays, cosmic rays, and neutrinos, and how do we detect them?*

Co-organizer, Southside Astronomy Program, 2025-26

- [Public talks](#) followed by guided night-sky viewing with telescopes for the community in the Southside of Chicago.

UChicago Astronomy Community Engagement Working Group, 2021 - present

- Organizing public outreach events including the [Southside Astro Art Contest](#)
- Won multiple UChicago *Inclusive Climate Grants* worth \$1000 to support community outreach efforts.

UChicago Climate Committee, 2024-2025

Adler Planetarium Volunteer, 2024 - present

- Monthly Astronomy Conversations volunteer
- Annual Operation Airlift mentor

Lifelong Learners Public Talk, 2024

- *The Solar Eclipse: coming to the U.S. on April 8, 2024*